

Chapter 10

Feature extraction

IMP302m

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10.1 Boundary feature descriptors

- The *length* of a boundary is one of its simplest descriptors. The number of pixels along a boundary is an approximation of its length.
- The *diameter* of a boundary B

$$diameter(B) = \max_{i,j} [D(p_i, p_j)]$$

D is a distance measure and p_i and p_j are points on the boundary

- Two extreme points that comprise the diameter is called the *major axis* (or *longest chord*) of the boundary

$$length_m = [(x_2 - x_1)^2 + (y_2 - y_1)^2]^{1/2} \quad angle_m = \tan^{-1} \left[\frac{y_2 - y_1}{x_2 - x_1} \right]$$

10.1 Boundary feature descriptors

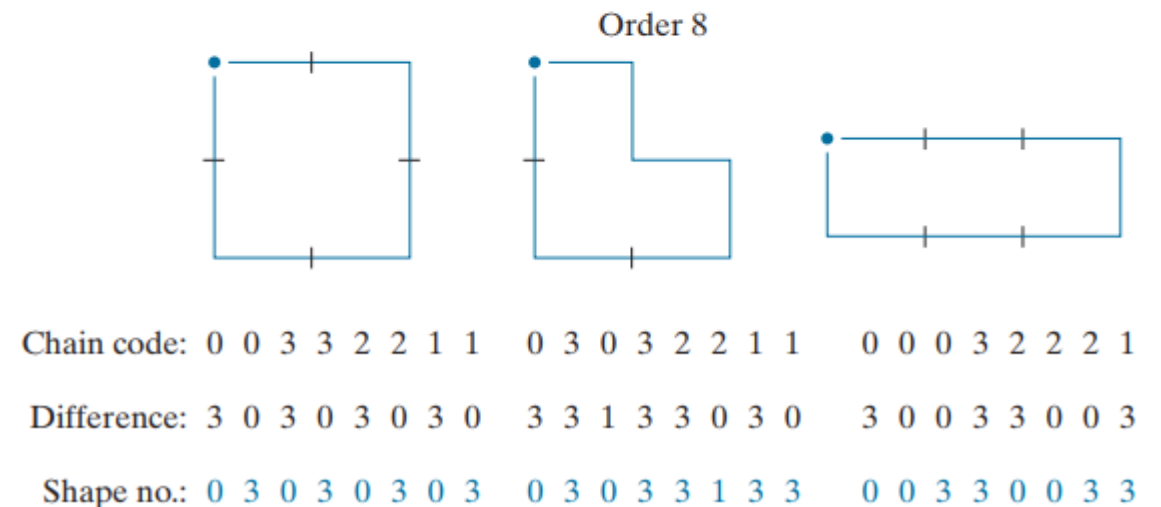
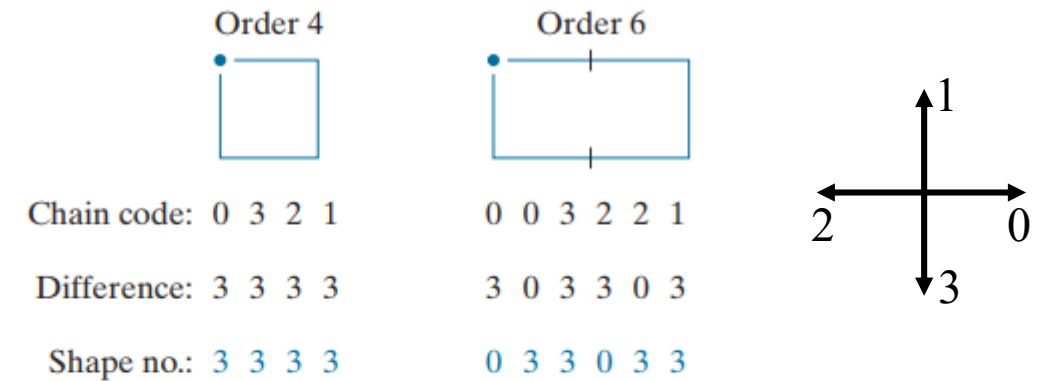
- The *minor axis* (also called the *longest perpendicular chord*) of a boundary is defined as the line perpendicular to the major axis
- The *curvature* of a boundary is defined as the rate of change of slope
 - formulated easily by expressing a boundary in the form of a slope chain code (SSC) defined as the sum of the absolute values of the chain elements

$$\tau = \sum_{i=1}^n |a_i|$$

- Example: Using slope chain codes to describe tortuosity

10.1 Boundary feature descriptors

- **Shape number** of a Freeman chain-coded boundary, based on the 4-directional code, is defined as the first difference of smallest magnitude



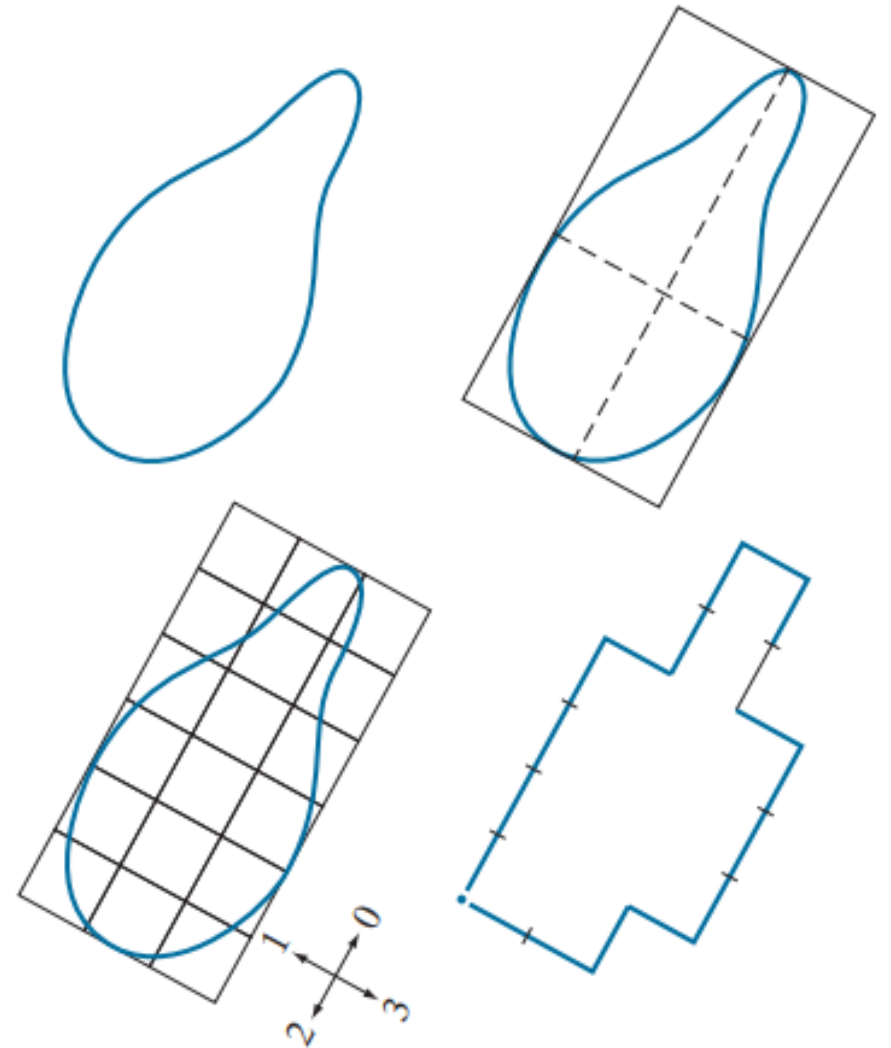
10.1 Boundary feature descriptors

- Shape number

Chain code: 0 0 0 0 3 0 0 3 2 2 3 2 2 2 1 2 1 1

Difference: 3 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0

Shape no.: 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0 3



10.1 Boundary feature descriptors

- **Fourier descriptors**
- the boundary itself can be represented as the sequence of coordinates $s(k) = [x(k), y(k)]$ for $k = 0, 1, 2, \dots, K-1$. Each coordinate pair can be treated as a complex number so that $s(k) = x(k) + jy(k)$
- Discrete Fourier transform (DFT) of $s(k)$ is

$$a(u) = \sum_{k=0}^{K-1} s(k) e^{-j2\pi uk/K}$$

The complex coefficients $a(u)$ are called the *Fourier descriptors* of the boundary

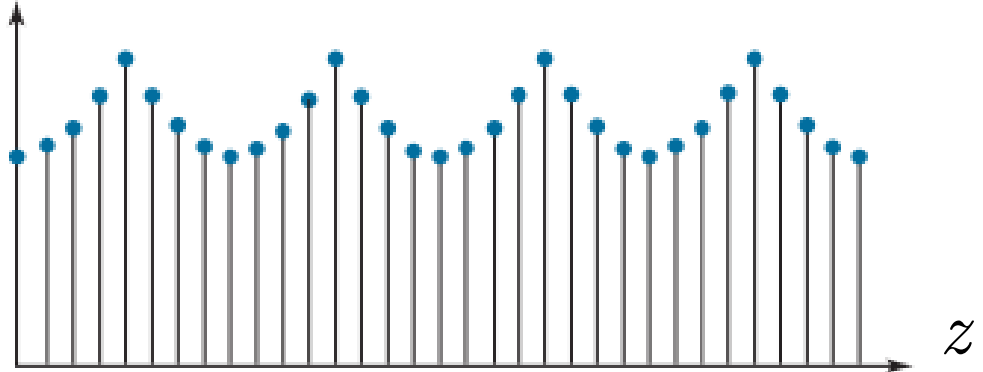
10.1 Boundary feature descriptors

- Discussion: How are Fourier descriptors if the boundary is transformed? Rotation, Translation, Scaling and Starting point difference

10.1 Boundary feature descriptors

- **Statistical moments**
- the *amplitude* of g as a discrete random variable z and form an amplitude histogram $p(z_i)$, $i=0,1,2,\dots,A-1$ where A is the number of discrete amplitude increments in which we divide the amplitude scale
- the n -th moment of z about its mean is

$$\mu_n(z) = \sum_{i=0}^{A-1} (z_i - m)^n p(z_i) \text{ where } m = \sum_{i=0}^{A-1} z_i p(z_i)$$



10.2 Region feature descriptors

- The *major* and *minor* axes of a region, as well as the idea of a *bounding box*, are as defined earlier for boundaries
- The *area* A of a region is defined as the number of pixels in the region
- The *perimeter* p of a region is the length of its boundary
- When area and perimeter are used as descriptors, they generally make sense only when they are normalized $\rightarrow$ *compactness* of a region

$$\text{compactness} = \frac{p^2}{A}$$

- *Circularity* (also called *roundness*)

$$\text{circularity} = \frac{4\pi A}{p^2}$$

10.2 Region feature descriptors

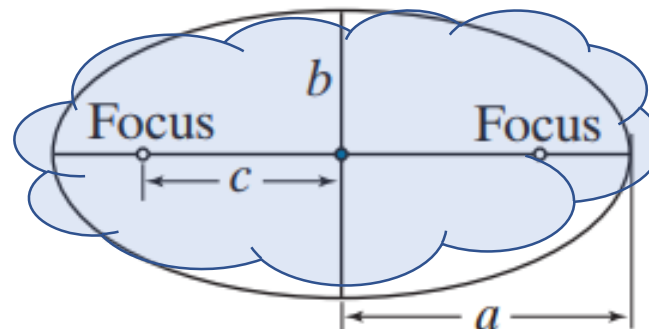
- *Effective diameter*

$$d_e = 2 \frac{A}{\pi}$$

- *Eccentricity* of a region relative to an ellipse as the eccentricity of an ellipse that has the same second central moments as the region

$$\text{eccentricity} = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - (b/a)^2} \quad a \geq b$$

$$c^2 = a^2 - b^2$$



10.2 Region feature descriptors

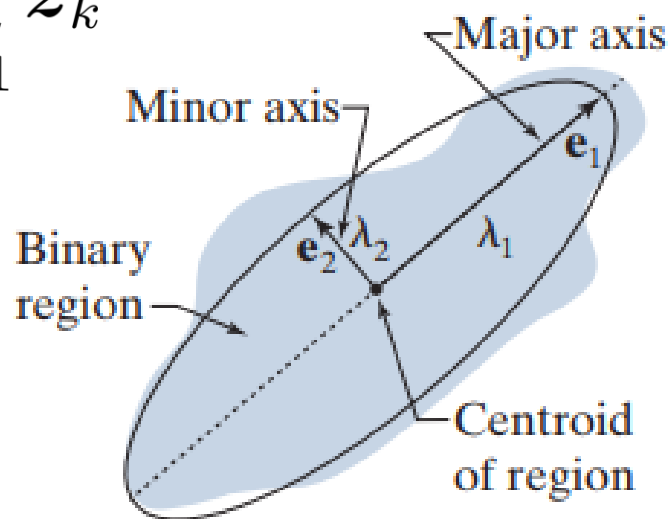
- *Eccentricity*: The view of principal axes

- Covariance matrix $\mathbf{C}$ of the data
$$\mathbf{C} = \frac{1}{K-1} \sum_{k=1}^K (\mathbf{z}_k - \bar{\mathbf{z}})(\mathbf{z}_k - \bar{\mathbf{z}})^T$$

where the mean vector $\bar{\mathbf{z}} = \frac{1}{K} \sum_{k=1}^K \mathbf{z}_k$

- $\mathbf{C} \rightarrow$ the eigenvectors and the eigenvalues

$$\begin{aligned} \text{eccentricity} &= \frac{\sqrt{\lambda_2^2 - \lambda_1^2}}{\lambda_2} \\ &= \sqrt{1 - (\lambda_1/\lambda_2)^2} \quad \lambda_2 \geq \lambda_1 \end{aligned}$$



$\mathbf{e}_1$, λ_1 and $\mathbf{e}_2$, λ_2 are the eigenvectors and corresponding eigenvalues of the covariance matrix of the coordinates of the region

10.2 Region feature descriptors

- Texture descriptors: provides measures of properties such as smoothness, coarseness, and regularity
- **Statistical Approaches**
- Let z be a random variable denoting intensity, $p(z_i)$, $i=0,1,2,\dots,A-1$ is the corresponding normalized histogram, where L is the number of distinct intensity levels, the n th moment of z about its mean is

$$\mu_n(z) = \sum_{i=0}^{L-1} (z_i - m)^n p(z_i) \text{ where } m = \sum_{i=0}^{L-1} z_i p(z_i)$$

- The second moment is particularly important in texture description. It is a measure of intensity contrast that can be used to establish descriptors of relative intensity smoothness

10.2 Region feature descriptors

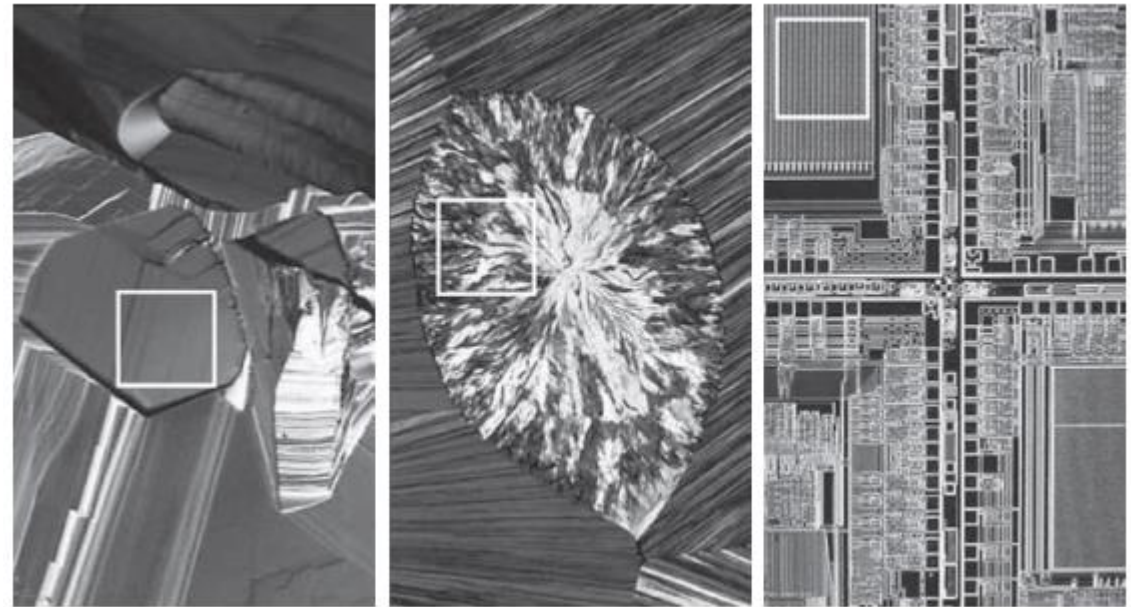
- Texture descriptors: provides measures of properties such as smoothness, coarseness, and regularity
- **Statistical Approaches**
- The third moment is a measure of the skewness of the histogram
- The fourth moment is a measure of its relative flatness.
- The fifth and higher moments are not so easily related to histogram shape, but they do provide further quantitative discrimination of texture content

- A measure of *uniformity*
$$U(z) = \sum_{i=0}^{L-1} p^2(z_i)$$

- A measure of *average entropy*
$$e(z) = - \sum_{i=0}^{L-1} p(z_i) \log_2 p(z_i)$$

10.2 Region feature descriptors

- Statistical texture measures for the subimages



Texture	Mean	Standard deviation	R (normalized)	3rd moment	Uniformity	Entropy
Smooth	82.64	11.79	0.002	-0.105	0.026	5.434
Coarse	143.56	74.63	0.079	-0.151	0.005	7.783
Regular	99.72	33.73	0.017	0.750	0.013	6.674

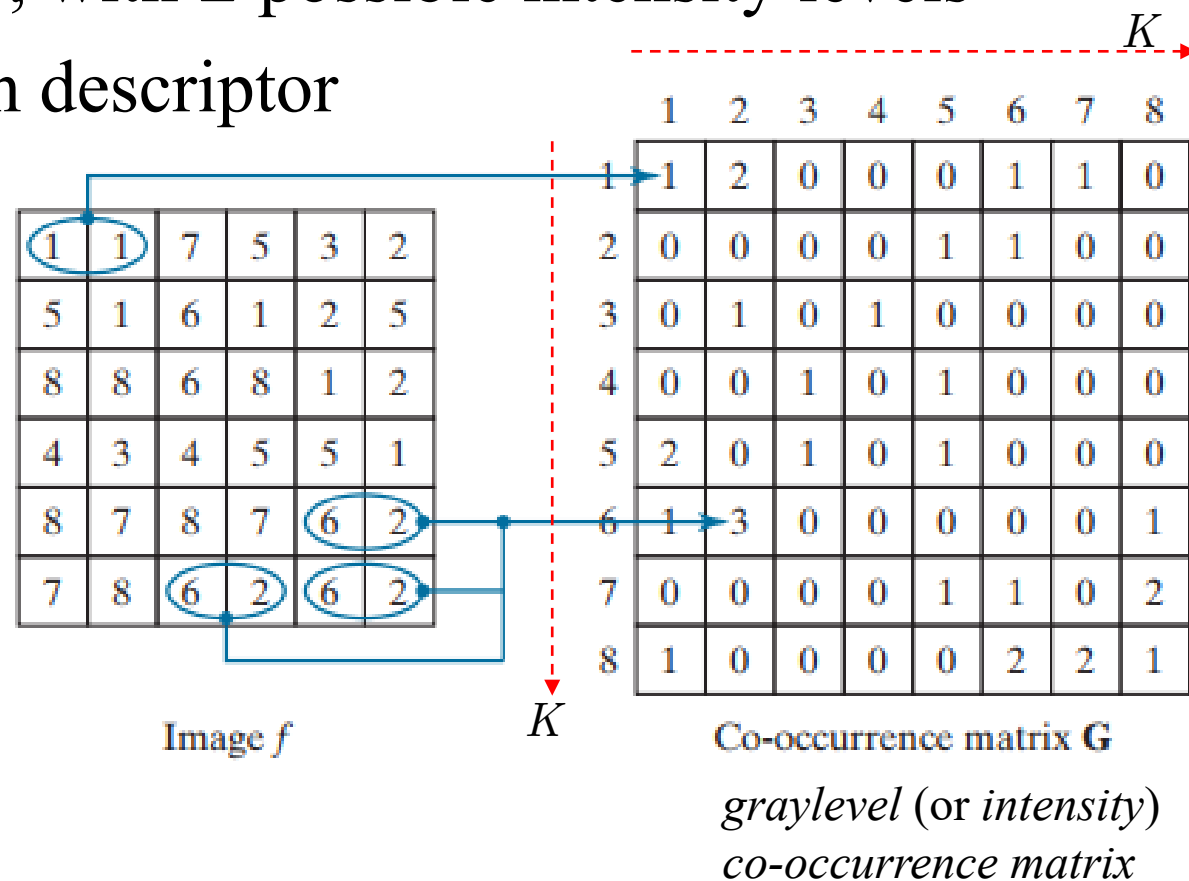
10.2 Region feature descriptors

- Let Q be an operator that defines the position of two pixels relative to each other, and consider an image, f , with L possible intensity levels
- The quantities used in the correlation descriptor

$$m_r = \sum_{i=1}^K i \sum_{j=1}^K p_{ij} \quad m_c = \sum_{j=1}^K j \sum_{i=1}^K p_{ij}$$

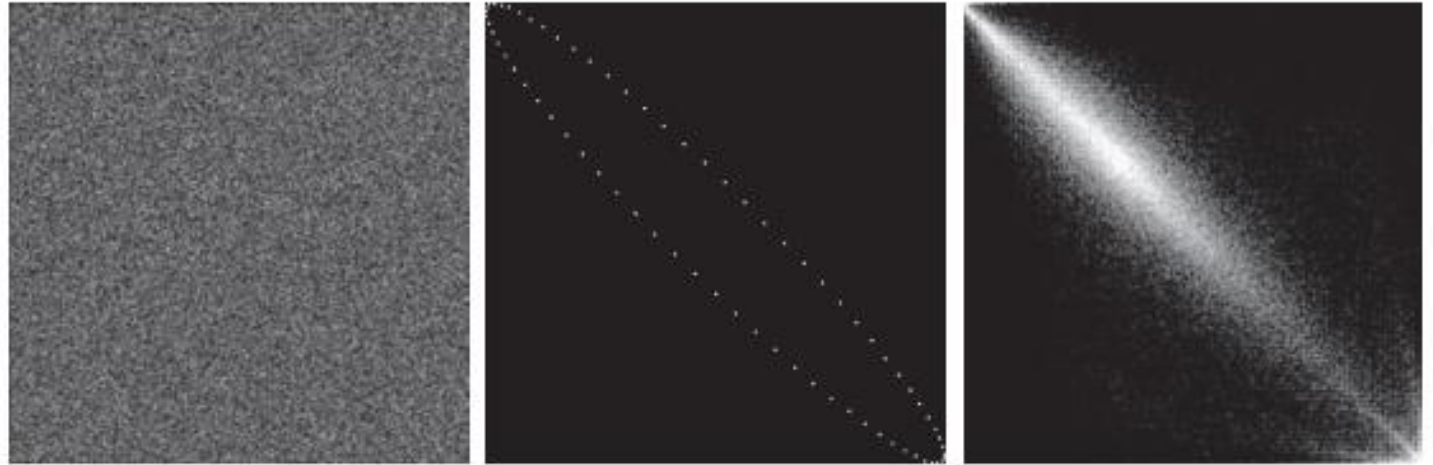
$$\sigma_r^2 = \sum_{i=1}^K (i - m_r)^2 \sum_{j=1}^K p_{ij}$$

$$\sigma_c^2 = \sum_{j=1}^K (j - m_c)^2 \sum_{i=1}^K p_{ij}$$



10.2 Region feature descriptors

- Descriptors evaluated using the co-occurrence matrices displayed as images



Normalized Co-occurrence Matrix	Maximum Probability	Correlation	Contrast	Uniformity	Homogeneity	Entropy
G_1/n_1	0.00006	-0.0005	10838	0.00002	0.0366	15.75
G_2/n_2	0.01500	0.9650	00570	0.01230	0.0824	06.43
G_3/n_3	0.06860	0.8798	01356	0.00480	0.2048	13.58

10.2 Region feature descriptors

Spectral Approaches

- Fourier spectrum is ideally suited for describing the directionality of periodic or semiperiodic 2-D patterns in an image
- These global texture patterns are easily distinguishable as concentrations of high-energy bursts in the spectrum
- Three features of the Fourier spectrum
 - 1) prominent peaks in the spectrum give the principal direction of the texture patterns
 - 2) the location of the peaks in the frequency plane gives the fundamental spatial period of the patterns
 - 3) eliminating any periodic components via filtering leaves nonperiodic image elements

10.2 Region feature descriptors

- **Spectral Approaches**

- Detection and interpretation of the spectrum features just mentioned often are simplified by expressing the spectrum in polar coordinates to yield a function $S(r, \theta)$

- For each direction θ , $S(r, \theta)$ may be considered a 1-D function $S_\theta(r)$
 - For each frequency r , $S(r, \theta)$ may be considered a 1-D function $S_r(\theta)$

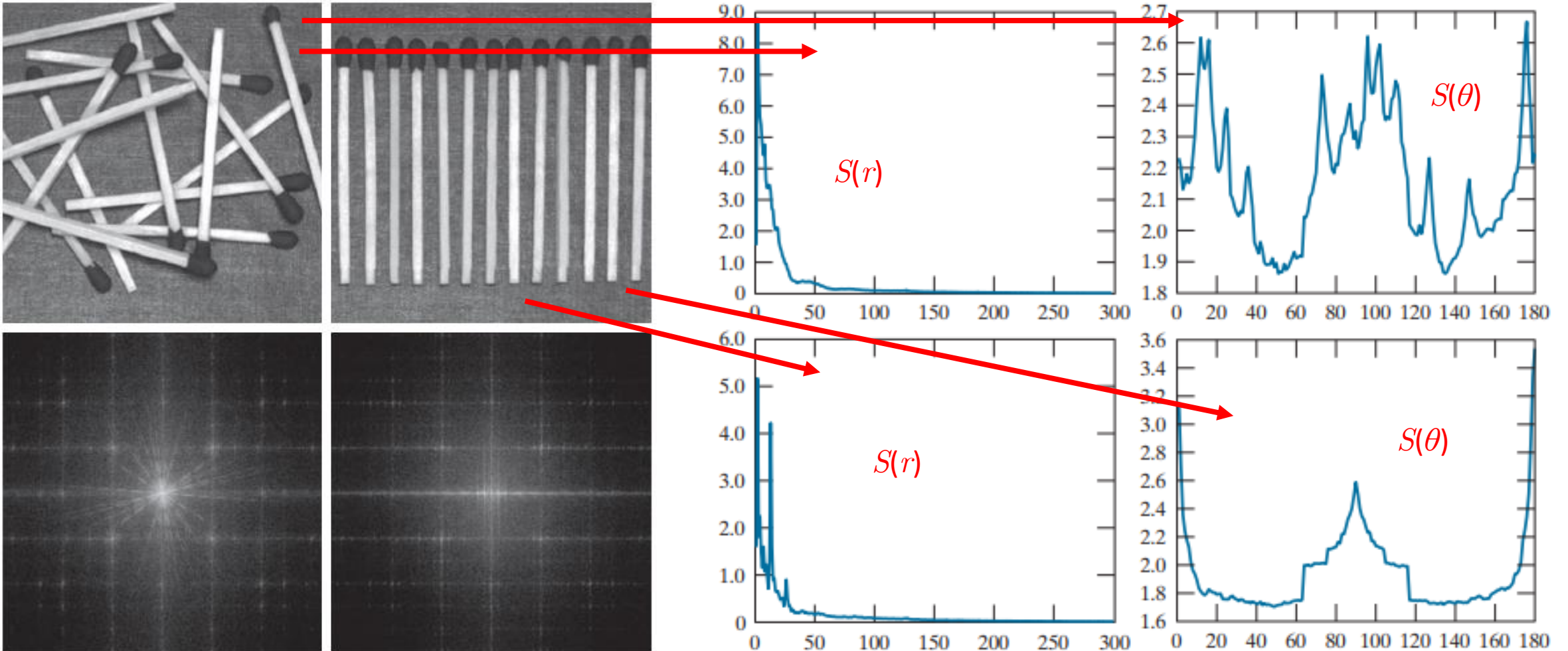
- for a fixed value of θ yields the behavior of the spectrum along a radial direction from the origin

$$S(\theta) = \sum_{r=1}^{R_0} S_r(\theta)$$

- for a fixed value of r yields the behavior along a circle centered on the origin.

$$S(r) = \sum_{\theta=0}^{\pi} S_\theta(r)$$

10.2 Region feature descriptors



10.2 Region feature descriptors

- **Moment invariants**

- The 2-D *moment* of order $(p+q)$ of an $M \times N$ digital image, $f(x, y)$, is

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p y^q f(x, y) \quad \text{where } p, q = 0, 1, \dots \text{ are integers}$$

- The corresponding *central moment* of order $(p+q)$

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y) \quad \text{where } \bar{x} = \frac{m_{10}}{m_{00}}; \bar{y} = \frac{m_{01}}{m_{00}}$$

- The *normalized central moment* of order $(p+q)$

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{\gamma}} \quad \text{where } \gamma = \frac{p+q}{2} + 1$$

10.2 Region feature descriptors

- A set of seven, 2-D *moment invariants* (to translation, scale change, mirroring (within a minus sign), and rotation) can be derived from the second and third normalized central moments

$$\phi_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12}) \left[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2 \right]$$

$$\phi_1 = \eta_{20} + \eta_{02} \quad + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03}) \left[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 \right]$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\phi_6 = (\eta_{20} - \eta_{02}) \left[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 \right]$$

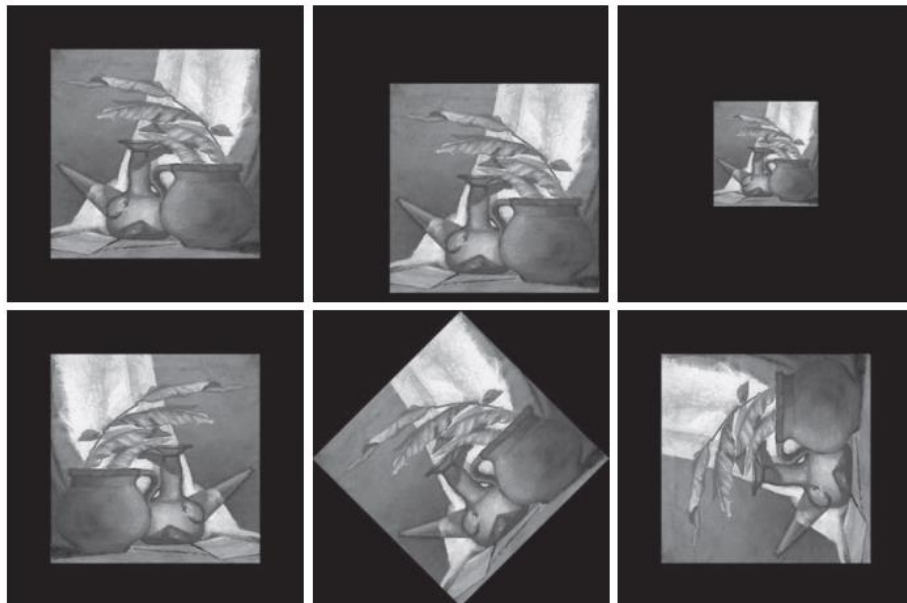
$$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \quad + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03})$$

$$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$$

$$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12}) \left[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2 \right] \\ + (3\eta_{12} - \eta_{30})(\eta_{21} + \eta_{03}) \left[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 \right]$$

10.2 Region feature descriptors

- Example



Moment Invariant	Original Image	Translated	Half Size	Mirrored	Rotated 45°	Rotated 90°
ϕ_1	2.8662	2.8662	2.8664	2.8662	2.8661	2.8662
ϕ_2	7.1265	7.1265	7.1257	7.1265	7.1266	7.1265
ϕ_3	10.4109	10.4109	10.4047	10.4109	10.4115	10.4109
ϕ_4	10.3742	10.3742	10.3719	10.3742	10.3742	10.3742
ϕ_5	21.3674	21.3674	21.3924	21.3674	21.3663	21.3674
ϕ_6	13.9417	13.9417	13.9383	13.9417	13.9417	13.9417
ϕ_7	-20.7809	-20.7809	-20.7724	20.7809	-20.7813	-20.7809

10.3 Principal Components as features

- n registered images, then the corresponding pixels at the same spatial location in all images can be arranged as an n -dimensional vector

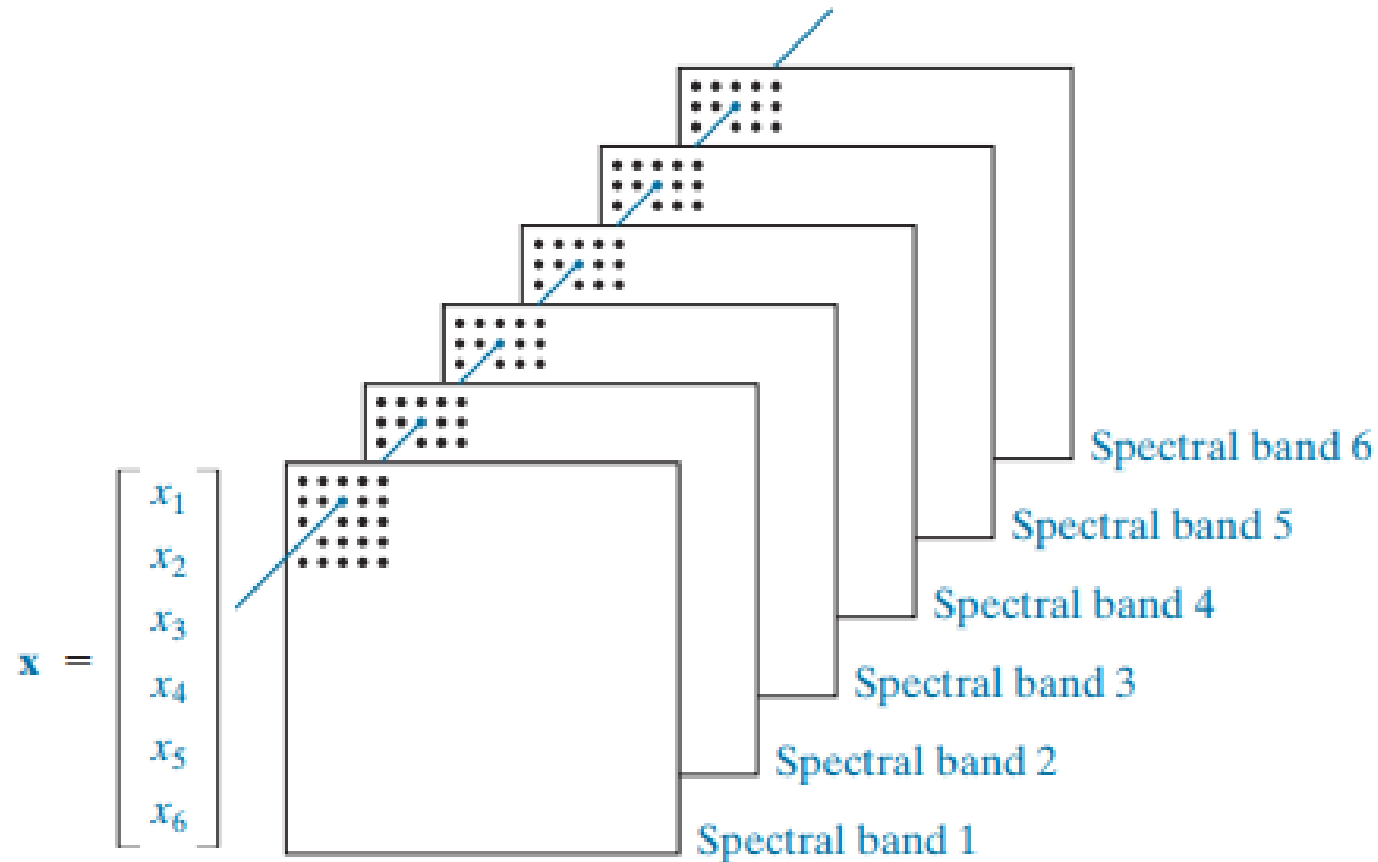
$$\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_n]^T$$

- Mean vector $\mathbf{m}_x = \mathbb{E} \{ \mathbf{x} \}$
- Covariance matrix $\mathbf{C}_x = \mathbb{E} \{ (\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^T \}$
- There are n eigenvectors and n eigenvalues derived from covariance matrix
- Suppose that we use $\mathbf{A}$ as a transformation matrix to map the $\mathbf{x}$'s into vectors

$$\mathbf{y} = \mathbf{A}(\mathbf{x} - \mathbf{y}_x)$$

10.3 Principal Components as features

- Example



10.4 Whole-image feature

- Principal feature detection methods
 - detecting corners
 - entire regions in an image

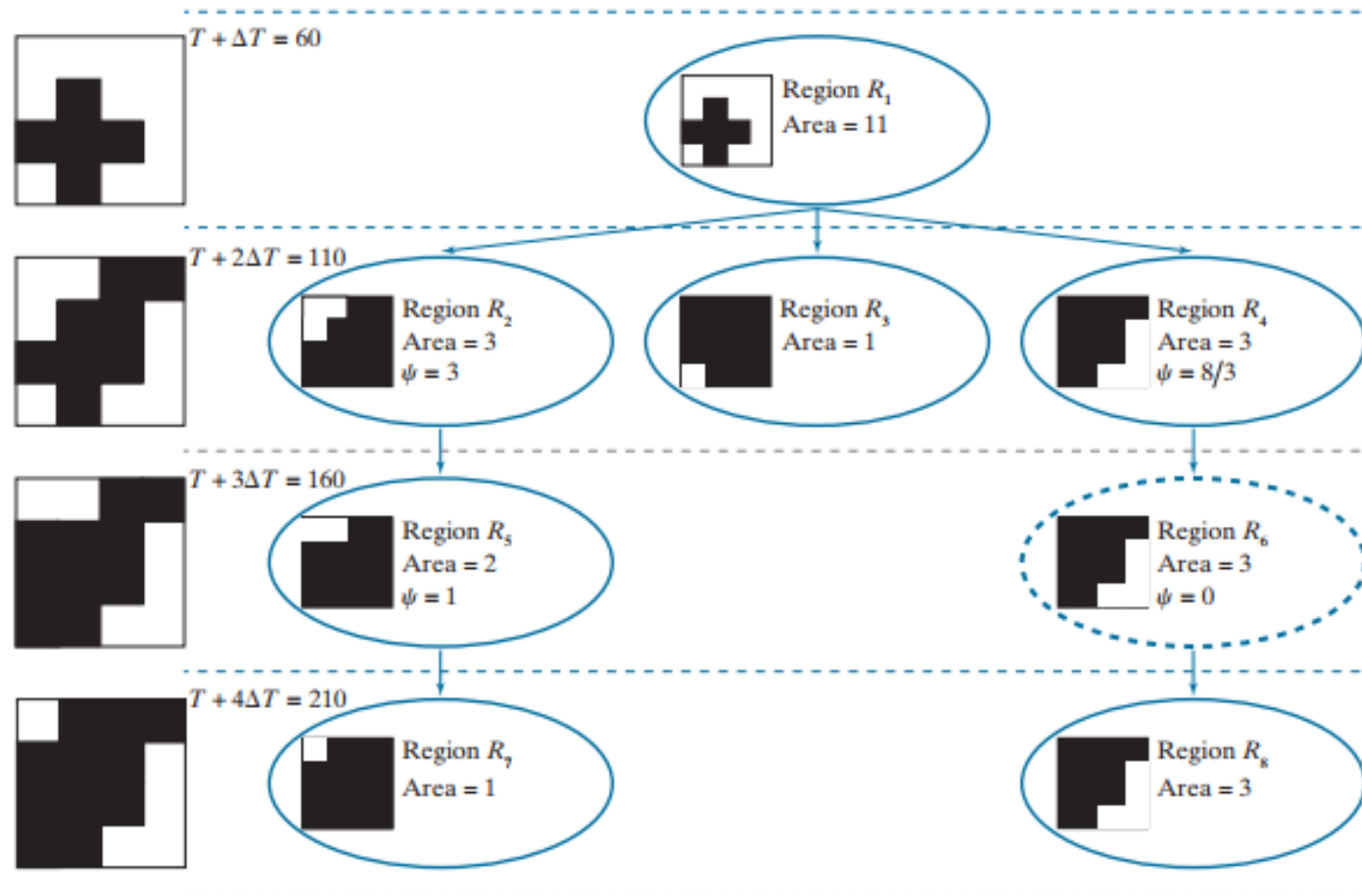
10.4 Whole-image feature

- Detecting corners
- The HARRIS-STEPHENS corner detector
- Remind from CPV301

10.4 Whole-image feature

- entire regions in an image
- Maximally Stable Extremal Regions (MSERs)**

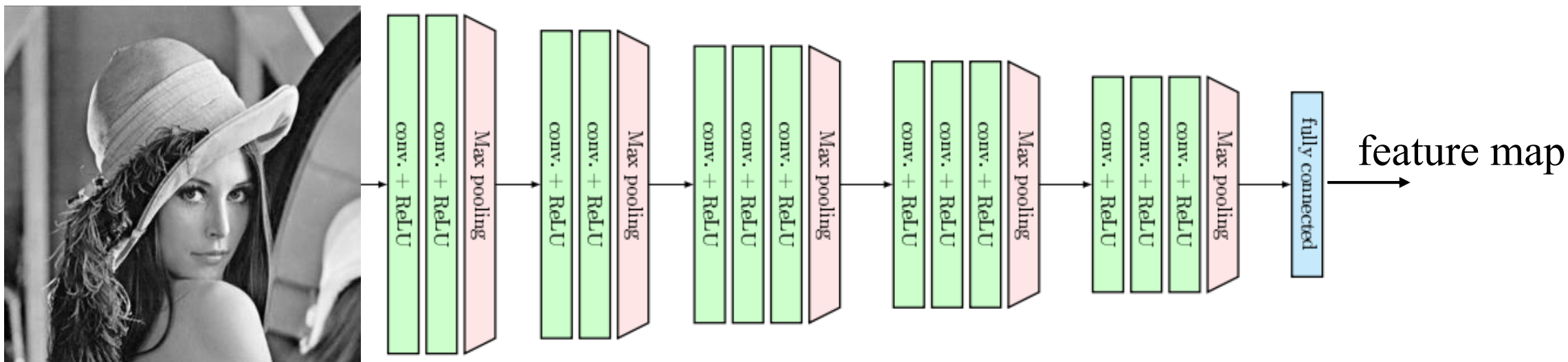
225	175	90	90
125	5	90	225
5	5	5	225
125	5	225	225



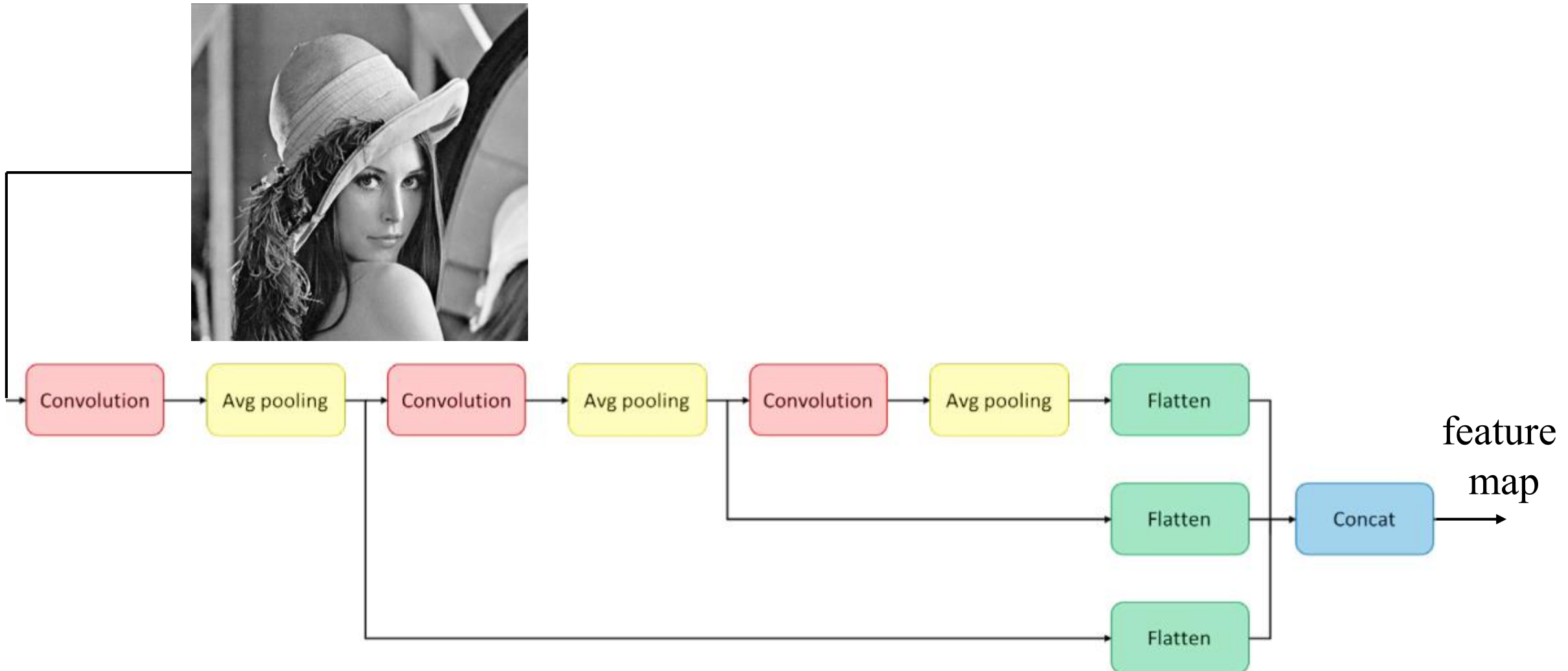
10.5 Machine learning based features

- Deep learning models can be used to extract features of images

VGG16



10.5 Machine learning based features



Summary

- Boundary feature descriptors
- Region feature descriptors
- Principal Components as features
- Whole-image feature
- Machine learning based features